ENTERPRISE GOOGLE CLOUD DATA PLATFORM BLUEPRINT

VOLUME 06 - Analytics Platform

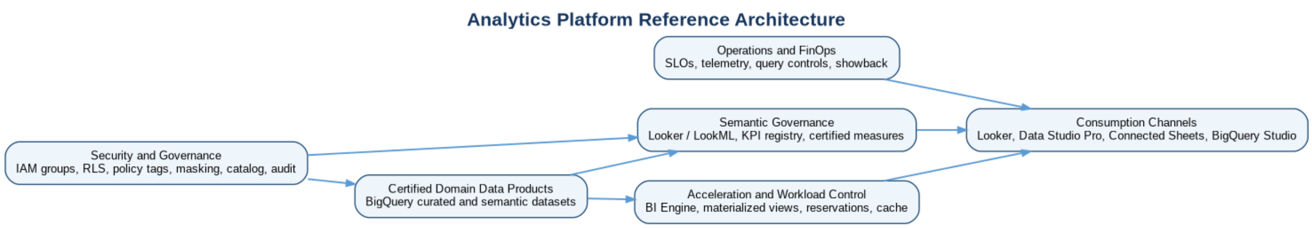

Figure 1 - Governed analytics from certified data products to trusted decisions

Fortune 100 Reference Architecture and Implementation Baseline

Architecture Baseline 1.0 | Publication date: 11 July 2026
Classification: Enterprise Reference Architecture

Document Control

This volume defines the governed enterprise analytics platform that consumes certified data products established in Volumes 03-05. It is a reusable reference architecture and implementation baseline, not a substitute for workload discovery, product licensing review, regulatory assessment, capacity testing or validation against current Google Cloud documentation.

Field	Value
Document	Volume 06 - Analytics Platform
Series	Enterprise Google Cloud Data Platform Blueprint
Version	1.0
Status	Detailed reference architecture for adaptation and implementation
Date	11 July 2026
Owner	Enterprise Analytics Architecture
Approvers	CDO, CIO, Enterprise Architecture, Data Governance, Security, Privacy, Platform Engineering, SRE and FinOps
Audience	Executives, analytics leaders, BI architects, analytics engineers, data-product teams, platform engineers, security, risk and audit
Review cycle	Quarterly and after material product, regulation, scale, licensing or architecture change
Primary dependencies	Volumes 01-05; Volume 08 Security; Volume 09 DevSecOps; Volume 10 FinOps; Volume 11 SRE; Volume 12 DR; Volume 13 Terraform
Scope	Analytics strategy, personas, semantic layer, consumption channels, performance, security, governance, delivery and operations

Current Product Terminology

Google Cloud product names and feature boundaries evolve. This volume uses Looker for the governed semantic and enterprise BI platform. It uses Data Studio Pro as the current Google Cloud naming for the managed enterprise version of the product historically known as Looker Studio Pro, while retaining “Looker Studio” where required by the master blueprint. BigQuery Studio refers to the integrated SQL, notebook and data-workspace experience. Knowledge Catalog is used for enterprise metadata and governance. BigQuery sharing is the current name for the capability formerly called Analytics Hub. Implementation teams must validate editions, licenses, locations, quotas, VPC Service Controls compatibility, APIs, IAM roles and Terraform provider arguments at delivery time.

Classification of guidance. Mandatory baseline means required unless an approved exception exists. Recommended pattern is the default for most workloads. Conditional pattern is used only when selection criteria are met. Future-state capability is included in the target architecture but is not required for initial production.

Table of Contents

1. Executive Summary

The enterprise analytics platform converts governed domain data products into consistent business decisions. Its core design separates authoritative data controls, semantic meaning and presentation. BigQuery remains the governed analytical data plane. Looker and LookML provide the primary enterprise semantic layer, reusable metrics, governed exploration and managed delivery. Data Studio Pro supports lightweight departmental visualization on certified sources. Connected Sheets supports controlled spreadsheet analysis. BigQuery Studio supports advanced SQL and notebook workflows for approved technical personas.

The architecture rejects two common failure modes: a single central BI team that becomes a delivery bottleneck, and uncontrolled self-service that creates thousands of conflicting metrics and extracts. The target operating model is federated. Domain teams own analytical meaning and product-specific semantic models. An enterprise analytics enablement team owns platform standards, reusable components, KPI certification workflows, delivery automation, shared capacity and support. Business owners remain accountable for KPI definitions and regulatory representations.

Security is enforced at the authoritative data layer using BigQuery IAM, row-level access policies, policy tags, column-level controls, dynamic masking, authorized views and regional boundaries. Looker roles, model sets, access filters and content permissions add a presentation and workflow layer but do not replace BigQuery security. Production analytics changes flow through Git, automated validation, peer review, controlled promotion and rollback. Query performance and cost are governed through model design, partitioning, clustering, materialized views, BI Engine, query cache, BigQuery reservations, workload assignments, concurrency testing and measurable unit economics.

Executive decision	Baseline position	Reason
Enterprise semantic layer	Looker/LookML is mandatory for certified enterprise metrics and governed exploration.	Central definitions with federated model ownership reduce contradictory KPIs and duplicated SQL.
Analytical data plane	BigQuery native curated and semantic datasets are the default.	Preserves the platform decisions inherited from Volume 03 and supports scale, security and workload control.
Self-service model	Tiered self-service with certified, governed and exploratory zones.	Balances delivery speed with trust, security and cost.
Secondary visualization	Data Studio Pro is conditional for lightweight, Google Workspace-oriented use cases.	Avoids forcing all use cases into one tool while preventing it from becoming a parallel semantic authority.
Spreadsheet analysis	Connected Sheets is conditional and bounded.	Preserves familiar analysis without uncontrolled exports or spreadsheet-based KPI ownership.
Capacity	Separate production BI from ad hoc workloads using reservations and assignments where justified.	Protects executive and regulatory workloads from noisy neighbors.
Security	BigQuery controls are authoritative; BI-layer controls are additive.	Prevents hidden fields or content folders from being mistaken for data security.
Delivery	Semantic models and production reports use Git and CI/CD.	Makes definitions reviewable, testable, auditable and reversible.

1.1 Expected business outcomes

Outcome	Target state	Illustrative measure
Trusted decisions	Priority executive and regulatory decisions use certified KPIs.	At least 90% of designated Tier 0 dashboards use certified measures.
Faster analysis	Analysts can explore certified products without waiting for bespoke extracts.	Median time from access approval to first governed analysis under one business day.
Metric consistency	One approved definition is reused across dashboards and regions.	No unresolved material variance between certified reports for the same KPI.
Scalable delivery	Domain teams deliver within common controls.	Standard semantic product promoted in two weeks after data-product readiness.
Cost transparency	Analytics consumption is attributed to domain, product, channel and environment.	At least 95% of BigQuery analytics spend attributable.
Operational reliability	Critical dashboards have SLOs, owners, runbooks and tested recovery.	Tier 0 and Tier 1 analytics pass production-readiness review.

Primary architecture outcome. A business user should be able to discover a certified KPI, understand its definition and owner, request access, analyze it through an approved channel, and trust that every result is secured, fresh, reconciled, observable and attributable to cost.

2. Inherited Context, Scope and Assumptions

This volume inherits the Fortune 100 enterprise scenario and the approved decisions from Volumes 01-05. It does not redefine the landing zone, data platform, pipeline patterns or data mesh. It translates those foundations into analytics-specific architecture, controls and operating practices.

Category	Inherited baseline	Analytics implication
Scale	10-50 PB historical estate, 50-250 TB/day, billions of events and thousands of users.	Semantic models, workload isolation, automated tests and cost controls must operate at enterprise scale.
Operating model	Central platform plus federated domain product teams.	Domain analytics engineers own product semantics; central teams own platform and cross-domain standards.
Data plane	BigQuery primary; Cloud Storage raw/replay/archive; BigLake/Iceberg selective.	Analytics reads certified BigQuery product interfaces by default and does not query raw data directly.
Governance	Knowledge Catalog, classification, lineage, data contracts and certification.	Every certified dashboard and metric links to a governed product and registered definition.
Security	Federated identity, group IAM, private-by-default, VPC Service Controls for sensitive workloads, selective CMEK.	Analytics identities and connections inherit geographic, classification and perimeter controls.
Regions	AMER, EMEA and APAC data planes with residency controls.	Models and content are location-aware; cross-region joins and extracts require approval.
Classifications	Public, Internal, Confidential, Restricted and Regulated.	Channel, sharing, export, embedding and caching controls derive from classification.
Delivery	Terraform and CI/CD authority for production platform and schema changes.	LookML, semantic SQL, capacity and access configuration are versioned and reviewed.
FinOps	Domain/product/environment attribution and showback.	BI capacity, query jobs, content and extracts carry accountable ownership.
SRE	Tiered SLOs, error budgets, monitoring, runbooks and recovery.	Dashboards and semantic services are treated as products with measurable reliability.

2.1 Scope boundaries

In scope	Out of scope or defined elsewhere
Analytics strategy, personas, governed self-service and channel selection.	Landing-zone resource hierarchy and network implementation - Volume 02.
Looker platform and semantic-model architecture.	Detailed data ingestion, storage and product interfaces - Volume 03.
KPI governance, certified measures and semantic standards.	Detailed Beam, Spark, Airflow and Dataform engineering - Volume 04.
Data Studio Pro, Connected Sheets, BigQuery Studio and notebooks.	Domain decomposition, product lifecycle and marketplace operating model - Volume 05.
Performance, caching, concurrency and analytics cost controls.	Complete zero-trust control mapping - Volume 08.
Analytics testing, deployment, SLOs and production readiness.	Enterprise CI/CD platform, FinOps, SRE and DR deep dives - Volumes 09-12.

2.2 Confirmed, assumed and open items

Type	Item
Confirmed requirement	Volume 6 must cover all analytics topics and representative artifacts listed in the master prompt.
Confirmed inherited decision	BigQuery is the primary governed analytical platform and Looker is the primary semantic analytics platform.
Architecture assumption	The enterprise licenses an appropriate Looker edition and can operate regional or approved central instances as required.
Architecture assumption	Priority data products expose analytics-ready curated or semantic interfaces with owners, contracts, quality and SLOs.
Mandatory control	Certified KPIs cannot be defined only in dashboard calculations or spreadsheets.
Recommended control	Production BI capacity is isolated from ad hoc and data-engineering workloads.

Type	Item
Conditional capability	Data Studio Pro, Connected Sheets, notebooks and embedded analytics are enabled by classification and use case.
Open architecture decision	Final Looker instance topology, edition and regional placement by legal entity.
Open architecture decision	Target capacity model, baseline slots, autoscaling limits and chargeback method.
Implementation dependency	Current product licensing, feature availability, quotas and regional support must be validated.

3. Analytics Strategy and Success Measures

The analytics strategy establishes a governed path from data products to decisions. It treats the semantic layer, reports and analytical workspaces as managed products rather than presentation files. The strategy is organized around six design commitments: business-led metrics, product-aligned semantics, tiered self-service, security at the data layer, reliability engineering and cost accountability.

ID	Strategic objective	Mandatory design response	Success measure
ANA-OBJ-01	One trusted metric language	Enterprise KPI registry and certified LookML measures.	Priority KPIs have one active certified definition per approved context.
ANA-OBJ-02	Governed self-service	Certified Explores, reusable content templates and controlled workspaces.	At least 80% of routine questions answered without new data extracts.
ANA-OBJ-03	Persona-fit channels	Looker, Data Studio Pro, Connected Sheets and BigQuery Studio selected by use case.	Every analytics asset records its approved channel and consumer persona.
ANA-OBJ-04	Performance at scale	Physical design, BI Engine, caching, reservations and concurrency tests.	Critical workloads meet p95 targets at 2x forecast peak.
ANA-OBJ-05	Security and privacy	Authoritative BigQuery controls plus BI-layer least privilege.	No critical analytics access finding; 100% restricted assets reviewed.
ANA-OBJ-06	Release quality	Git, automated validation, reconciliation, content testing and rollback.	No unreviewed production semantic-model changes.
ANA-OBJ-07	Operational reliability	SLOs, monitoring, runbooks, ownership and error budgets.	Tier 0/1 assets have active SLOs and on-call routing.
ANA-OBJ-08	Cost accountability	Workload attribution, budgets, unit metrics and optimization reviews.	95% of analytics compute attributable to domain/product/channel.

3.1 Analytics principles translated into controls

Principle	Concrete architecture control
Certified before broad reuse	A metric must have an owner, formula, grain, source product, reconciliation and version before certification.
Explore before extract	Users receive governed query access before approval for copies or flat-file exports.
Data-layer security is authoritative	RLS, policy tags, masking and authorized views protect data even when a BI asset is misconfigured.
Definitions as code	LookML and semantic SQL are versioned, tested and promoted through CI/CD.
Performance is designed, not cached away	Models use correct grain, selective joins, partition pruning and aggregate strategies before adding capacity.
Workloads have owners	Every model, dashboard, schedule, notebook and capacity assignment maps to a product owner and support tier.
Regionality is explicit	Model connections, content, extracts and sharing follow the regional product boundary.
Self-service has expiry and hygiene	Personal content, temporary tables, notebooks and extracts have ownership and lifecycle rules.

4. Analytics Personas and Consumption Channels

A single analytics tool cannot serve every persona safely or economically. The platform therefore offers a small, governed portfolio. Channel selection is based on decision criticality, semantic reuse, interactivity, collaboration pattern, development skill, embedding needs, classification and operational ownership.

Persona	Primary needs	Default channel	Guardrails
Executive / board consumer	Stable KPIs, narrative, mobile, scheduled delivery, alerts.	Looker governed dashboard.	Certified measures only; no ad hoc calculations on Tier 0 content.
Business leader	Team performance, drill paths, variance explanation.	Looker dashboard and Explore.	Product-aligned model; access by geography and function.
Business analyst	Interactive exploration and reusable team reports.	Looker Explore; Data Studio Pro conditionally.	Certified sources, content folders, query limits and review before broad publication.
Analytics engineer	Semantic models, metrics, tests and performance engineering.	LookML, BigQuery SQL and Git.	CI/CD, peer review, service identity and release ownership.
Power analyst	Complex SQL, notebooks and spreadsheet pivoting.	BigQuery Studio; Connected Sheets conditionally.	Approved project, maximum bytes, expiry, no direct raw access.
Regulatory / finance controller	Controlled report, evidence, reproducibility and sign-off.	Looker controlled report and approved export process.	Locked definition, reconciliation, retention, reviewer evidence and release freeze.
Operational user	Near-real-time status and exception workflow.	Looker operational dashboard or embedded analytics.	Freshness SLO, alert routing, bounded query load and failure fallback.
External customer / partner	Tenant-specific analytics inside an application.	Looker embedded analytics.	Server-generated signed or cookieless embed, tenant entitlements, RLS and contract limits.
Data scientist	Exploration and feature understanding.	BigQuery Studio notebooks / Vertex AI workbench pattern in Volume 7.	Certified product access, isolated compute, reproducibility and sensitive-data controls.

4.1 Channel selection matrix

Requirement	Looker	Data Studio Pro	Connected Sheets	BigQuery Studio
Certified enterprise KPIs	Primary	Consume only	Consume only	Validation/analysis
Governed exploration	Primary	Limited	Limited	SQL-driven
Lightweight visualization	Supported	Primary	Basic	Not primary
Spreadsheet workflow	Export/linked use	No	Primary	No
Advanced SQL/notebook	Limited	No	No	Primary
Embedded analytics	Primary	Not baseline	No	Not baseline
Regulatory reporting	Primary with controls	Conditional	Not authoritative	Support/reconciliation
Restricted data	Supported with layered controls	Conditional	Strongly restricted	Conditional
Version-controlled semantics	LookML	Report source only	No	SQL/notebook assets
Large concurrency	Designed and capacity-tested	Connector dependent	Connector dependent	Workload-isolated

Mandatory channel rule. Data Studio Pro reports, Connected Sheets workbooks and BigQuery Studio notebooks may consume certified measures, but they are not the enterprise system of record for KPI definitions.

5. Target Analytics Architecture

The target architecture separates five planes. The data-product plane supplies certified, regionally governed datasets. The semantic plane converts product schemas into business language and governed metrics. The consumption plane provides persona-appropriate experiences. The acceleration plane manages performance and capacity. The control plane provides identity, governance, CI/CD, observability and FinOps.

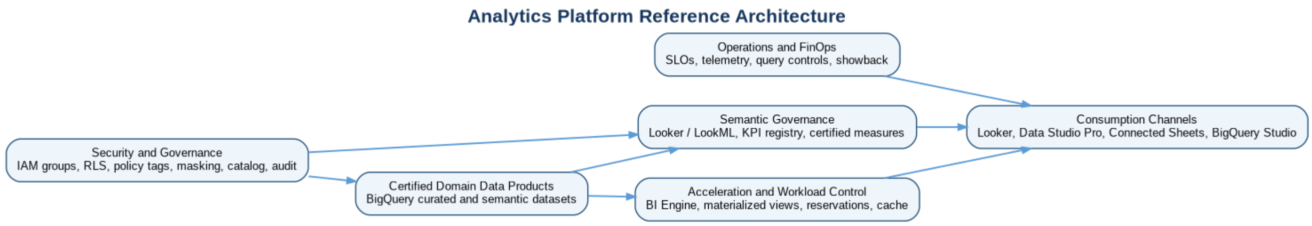

Figure 2 - Enterprise analytics platform reference architecture

Plane	Core components	Ownership	Key controls
Data-product plane	BigQuery curated/semantic datasets, authorized views, materialized views.	Domain data-product teams.	Contracts, quality, lineage, classification, retention and SLOs.
Semantic plane	Looker instance, LookML projects, Explores, metric definitions, model documentation.	Federated analytics engineers with enterprise standards.	Git, validation, certification, model sets and release management.
Consumption plane	Looker dashboards/alerts/embed, Data Studio Pro, Connected Sheets, BigQuery Studio.	Content owners and business consumers.	Folder governance, publication tiers, sharing, export and lifecycle.
Acceleration plane	BigQuery reservations, assignments, BI Engine, cache and aggregate structures.	Analytics platform and FinOps.	Capacity model, concurrency test, workload priorities and cost attribution.
Control plane	Identity groups, IAM, policy tags, masking, VPC SC, KMS, logging, monitoring, catalog and CI/CD.	Platform, security, governance, SRE and FinOps.	Least privilege, policy as code, audit, SLOs, evidence and exception expiry.

5.1 Logical data flow

Step	Flow	Control point
1	Domain pipelines publish a certified analytics interface in BigQuery.	Product certification confirms contract, quality, freshness, lineage and classification.
2	Looker connects with an approved service identity to region-compatible datasets.	Connection scopes, OAuth/service-account controls, private connectivity where supported and audit logs.
3	LookML maps technical schemas to reusable dimensions, joins, measures and access rules.	Code review, tests, KPI registry link, naming and fanout validation.
4	Consumers access content through persona and model-set roles.	Group membership, least privilege, folder/content permissions and product entitlement.
5	BigQuery executes under workload assignments and data-layer security.	RLS, policy tags, masking, authorized views, reservation and query controls.
6	Results may use query cache, BI Engine or aggregate structures when eligible.	Classification-aware caching, freshness monitoring and performance validation.
7	Telemetry is correlated across content, Looker queries, BigQuery jobs, costs and SLOs.	Owner routing, incident response, optimization and compliance evidence.

5.2 Physical deployment pattern

The recommended physical pattern uses approved analytics projects per environment and geography, separate from domain data projects. A Looker instance or approved topology connects only to allowed regional data products. Production connections are distinct from development and test. BigQuery reservations are assigned to analytics consumer projects or folders according to workload class. Restricted and regulated products remain within approved service perimeters and are not copied into a global BI mart merely for convenience.

Resource	Example	Design note
Analytics project	acme-ana-platform-prod-amer-01	Hosts analytics-specific services, telemetry and controlled integrations.
Domain data project	acme-data-sales-prod-amer-01	Owns certified data products and authoritative data controls.
Semantic dataset	sales_semantic	Stable product-aligned views/tables optimized for analytics.
Looker connection	bq-prod-amer-certified	Read/query access to approved products only; no raw-zone access.

Resource	Example	Design note
Reservation	bqrs-analytics-prod-amer	Capacity boundary for production analytical queries.
IAM group	gcp-sales-analytics-amer@acme.example	Persona/domain/geography membership; no individual grants.
Service account	sa-looker-query-prod-amer	Workload identity for approved Looker connection where applicable.
Cost labels/tags	domain=sales; product=order-performance; env=prod	Required on projects, assets and supporting deployment metadata.

6. Governed Self-Service Analytics

Governed self-service is not unrestricted access to every table. It is the ability to answer new questions quickly using pre-approved data products, reusable semantics, bounded workspaces and automated controls. The model uses four publication tiers.

Tier	Purpose	Who can publish	Controls	Lifecycle
Certified enterprise	Executive, regulatory and cross-domain decisions.	Authorized analytics engineers after owner and governance approval.	Certified KPI, reconciliation, security review, SLO, CI/CD and change record.	Managed release; formal deprecation.
Governed domain	Reusable domain analysis and operational management.	Domain analytics team.	Product source, code review, tests, owner, documentation and performance review.	Quarterly content review.
Team self-service	Departmental analysis on certified sources.	Delegated creators.	Approved channel, folder, source, sharing boundary, naming and expiry.	Review after inactivity; promote or retire.
Personal exploratory	Individual discovery and prototypes.	Named users.	Nonproduction billing project, byte limits, no broad sharing, temporary outputs and expiry.	Automatic cleanup.

6.1 Self-service paved road

- Discover a certified product and KPI in the enterprise catalog.
- Request access through the standard workflow with purpose, duration and geography.
- Use an approved Looker Explore, Data Studio Pro data source, Connected Sheets connection or BigQuery Studio workspace.
- Create content in a personal or team development area.
- Validate data, performance, security and business meaning.
- Promote reusable content through the appropriate certification path.
- Monitor usage, freshness, failures and cost; retire content that no longer has an owner or consumer.

6.2 Anti-patterns

Anti-pattern	Why it fails	Required response
Dashboard-defined enterprise metrics	Calculations diverge across reports and are difficult to audit.	Move approved logic to semantic model or certified data product.
Direct analysis of raw/bronze data	Bypasses contracts, quality, privacy and stable business meaning.	Use curated/semantic interface or approved engineering exception.
Shared service-account credentials	Removes user attribution and expands blast radius.	Use governed connection identities and user-level authorization patterns.
Spreadsheet extracts as system of record	Creates stale, uncontrolled copies and hidden business logic.	Use live governed connectivity and document any approved export.
One global model with every join	Creates fanout, permission complexity and slow development.	Use product-aligned models with shared conformed components.
Cache as the only performance strategy	Masks bad SQL, stale semantics and capacity problems.	Correct model and physical design first; then apply acceleration.
Production content owned by individuals	Content becomes orphaned during role changes.	Assign business and technical owners through groups and product metadata.

7. Looker Architecture

Looker is the primary governed enterprise analytics platform because it combines a version-controlled semantic modeling layer, governed exploration, dashboarding, scheduling, alerts, APIs and embedding. The architecture uses Looker as a semantic and consumption service, not as a replacement for BigQuery governance or data engineering.

Component	Enterprise role	Baseline decision
Looker instance topology	Provides regional or approved central analytics runtime.	Separate production from development/test; select topology based on residency, latency, licensing and operating model.
BigQuery connections	Provide governed query path to certified products.	Separate by environment/geography and materially different trust boundary; no wildcard raw access.
LookML projects	Organize semantic code by product/domain and reusable enterprise components.	Git-backed with branch development, code owners, CI and release tags.
Models and model sets	Bound data access and role scope.	Product-aligned models; role = permission set plus model set.
Explores	Govern user exploration and join paths.	Certified Explores expose explicit fields, valid joins, default filters and documentation.
Folders/content	Organize dashboards, Looks and schedules.	Separate certified, domain, team and personal content; group ownership.
API/embed	Support automation and application analytics.	Dedicated clients, server-side token generation, tenant entitlements, rate and audit controls.
Administration	Manage users, roles, connections, content and system activity.	Privileged access is time-bound, audited and separated from content development.

7.1 Instance and regional strategy

The final instance count is an open decision because Looker deployment options, licensing and supported locations must be validated. The baseline is to avoid a single unconstrained global instance when regulated data, latency or administrative separation requires regional boundaries. Each instance or connection must have an explicit geography, data classification ceiling, recovery tier and owner. Cross-region content should consume pre-approved aggregated products rather than perform uncontrolled cross-region queries.

7.2 Connection security

Control	Requirement
Identity	Use an approved workload identity or connection method. Avoid user-managed service-account keys.
Scope	Grant BigQuery job creation and dataset access only where required. Prefer views/product interfaces over table-wide project grants.
Network	Use private connectivity patterns supported by the selected Looker deployment; validate PSC/VPC SC compatibility and egress paths.
Encryption	Use Google-managed encryption by default and CMEK where inherited classification/contract requires and product support is confirmed.
Audit	Enable and retain relevant Looker activity, BigQuery job and Data Access logs; correlate user/content/query identifiers.
Separation	Development connections cannot write or query production except through explicitly approved read-only validation paths.
Secrets	Store connection secrets in approved secret management; rotate and test failure behavior.

7.3 Content architecture

Content area	Purpose	Publishing authority	Minimum evidence
Enterprise certified	Board, executive, regulatory and cross-domain content.	Central/delegated certification team.	KPI approval, reconciliation, SLO, security, performance and release evidence.
Domain certified	Reusable domain management content.	Domain analytics owner.	Product lineage, model tests, owner, documentation and query test.
Team shared	Collaborative analysis.	Delegated team creators.	Certified source, naming, sharing check and owner.
Personal	Exploration and drafts.	Individual creator.	Automatic lifecycle; no external sharing.
Embedded	Application-facing analytical experience.	Application product owner plus security review.	Tenant model, authorization test, load test, monitoring and support.

8. LookML, Semantic Modeling and Metrics Layer

The semantic layer is the controlled translation between physical data products and business language. It defines dimensions, measures, join relationships, default filters, drill paths, descriptions, formats and access behavior. It must preserve the grain and contract of source products rather than conceal structural defects.

8.1 Semantic model structure

Layer	Contents	Standard
Connection	Environment/geography-specific BigQuery connection.	No business logic; least privilege; explicit ownership.
Base views	Direct mappings to certified tables/views.	Stable field names, primary keys, descriptions and hidden technical fields.
Derived/aggregate views	Approved persistent or native aggregate structures.	Use only with clear refresh, lineage, cost and failure behavior.
Explores	Consumer-oriented join graph and field exposure.	Correct relationships, bounded joins, useful defaults and query-safety controls.
Measures	Reusable aggregations and business calculations.	Certified when enterprise-critical; never double-count across fanout.
Dimensions	Business attributes and time groups.	Conformed naming, formats, glossary mapping and classification.
Access constructs	Access grants/filters and model scopes.	Additive to authoritative BigQuery security; tested for deny and allow paths.
Content	Dashboards, Looks and alerts.	Consume model fields rather than duplicate calculations.

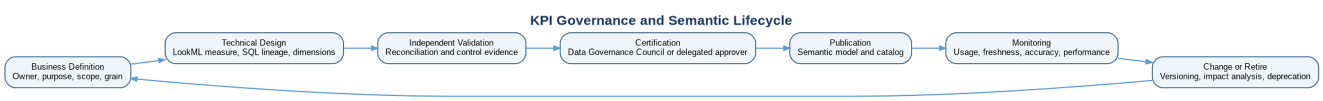

Figure 3 - KPI governance and semantic-model lifecycle

8.2 Semantic-layer standards

Standard	Mandatory rule	Acceptance evidence
Grain	Every view documents one business grain and stable key.	Model documentation and uniqueness test.
Join cardinality	Every join declares and validates its relationship.	Fanout test with representative data.
Measure additivity	Measures document additive, semi-additive or non-additive behavior.	KPI record and period/dimension tests.
Time semantics	Business date, event time, processing time and timezone are explicit.	Field description and boundary tests.

Standard	Mandatory rule	Acceptance evidence
Null/unknown handling	Unknown, not applicable and missing are distinguished where material.	Test cases and dashboard display standard.
Currency/unit	Currency, unit, conversion rate and effective date are explicit.	Reconciliation and format test.
Naming	Business-friendly singular dimensions and clear measure nouns.	Lint/peer review.
Descriptions	Certified fields include definition, owner, source and limitations.	Metadata completeness check.
Security	Sensitive fields remain protected in BigQuery and are not merely hidden.	Negative access test.
Versioning	Breaking changes use versioning, impact analysis and migration window.	Release record and content validation.

8.3 Metrics-layer design

The metrics layer is implemented through certified LookML measures backed by certified product fields or governed semantic SQL. Metrics are organized by business capability and owner. Cross-domain metrics use conformed dimensions and explicit contracts. A metric may have contextual variants, such as statutory and management revenue, but each variant must be named, owned and certified; the platform does not force false uniformity.

Metric element	Required metadata
Identity	KPI ID, name, aliases and version.
Meaning	Business definition, purpose, exclusions and decision supported.
Computation	Formula, grain, aggregation behavior and rounding.
Data	Source products, fields, lineage and effective date.
Ownership	Business owner, steward, technical owner and approver.
Control	Classification, reconciliation, quality threshold and freshness.
Presentation	Format, default filters, drill path and warning text.
Lifecycle	Status, review date, change policy, consumers and deprecation date.

8.4 Representative LookML

The package includes a complete compact LookML example under examples/looker. The excerpt below demonstrates a certified measure, product-aligned table reference and access-controlled field. It is illustrative production-oriented code and must be validated against the target Looker release and connection.

```
view: order {
  sql_table_name: `acme-data-sales-prod-amer-01.sales_semantic.order_fact` ;;
  dimension: order_id { primary_key: yes type: string sql: ${TABLE}.order_id ;; }
  dimension_group: order { type: time timeframes: [date, week, month, quarter, year] sql: ${TABLE}.order_ts ;; }
  dimension: customer_id { hidden: yes type: string sql: ${TABLE}.customer_id ;; }
  dimension: region_code { type: string sql: ${TABLE}.region_code ;; }
  measure: order_count { type: count_distinct sql: ${order_id} ;; value_format_name: decimal_0 }
  measure: gross_revenue { type: sum sql: ${TABLE}.gross_revenue_amount ;; value_format_name: usd }
  measure: net_revenue {
    type: sum sql: ${TABLE}.net_revenue_amount ;; value_format_name: usd
    description: "Certified net revenue after approved deductions."
    tags: ["certified-kpi", "owner:finance-performance"]
  }
  measure: restricted_margin {
    type: sum sql: ${TABLE}.margin_amount ;; value_format_name: usd
    required_access_grants: [finance_restricted]
  }
  set: detail { fields: [order_id, order_date, region_code, net_revenue] }
}
```

9. KPI Governance and Certified Measures

KPI governance is a business-control process supported by code and metadata. The Data Governance Council delegates day-to-day approval to accountable KPI councils or domain owners while retaining policy authority. Certification means that a measure is approved for a stated context; it does not imply that one formula is valid for every legal entity, accounting basis or operational decision.

Lifecycle gate	Accountable role	Required evidence	Automation
Propose	Business owner	Purpose, definition, decisions, consumers and context.	KPI intake template and unique ID.
Design	Analytics engineer and steward	Formula, grain, source, dimensions, classification and limitations.	Schema/model validation.
Validate	Independent controller or subject expert	Reconciliation, sample cases, edge cases and tolerance.	Automated tests and comparison report.
Approve	Delegated KPI authority	Owner sign-off, regulatory/privacy review where needed.	Workflow record and catalog status.
Publish	Analytics product owner	LookML release, documentation, dashboard impact and SLO.	CI/CD and catalog update.
Operate	Product owner and SRE	Freshness, accuracy, usage, cost and incident response.	Monitoring and scorecard.
Change/retire	Business owner	Impact analysis, consumer notice and migration plan.	Content validation and deprecation tracking.

9.1 Certified-measure controls

- A certified measure has exactly one active owner and an explicit approval scope.
- The formula is implemented in the governed semantic layer or certified product, not copied into each dashboard.
- Reconciliation uses an independent control total for financial, regulatory and Tier 0 measures.
- The semantic model and source product versions are recorded.
- Breaking changes require consumer impact analysis, parallel validation and a migration window.
- Dashboard tiles display certification status and data freshness where decision risk justifies it.
- Certification can be suspended after unresolved quality, control or ownership failure.

9.2 KPI definition template

Field	Example
KPI ID	KPI-SALES-001
Name	Net Revenue
Business definition	Recognized revenue after approved deductions.
Context	Management reporting; not statutory revenue unless explicitly stated.
Owner	VP Finance Performance
Steward	Sales Analytics Steward
Grain	Day x legal entity x region x channel
Formula	SUM(net_revenue_amount)
Source product	sales/order-performance/v2
Semantic field	order.net_revenue
Classification	Confidential
Freshness	Available by 06:00 local business time
Accuracy	Reconciles to approved finance control total within tolerance
Version / review	2.1 / quarterly

10. Data Studio Pro and Connected Sheets

10.1 Data Studio Pro (Looker Studio)

Data Studio Pro is a conditional enterprise visualization channel for lightweight reports, Google Workspace collaboration and departmental use cases that do not require Looker’s full governed semantic workflow. It must use certified BigQuery views, approved Looker semantic connectors where supported, or managed data sources. It may not become an alternate metric authority.

Use when	Do not use when
A lightweight report can be built on an existing certified source.	The use case is an executive, regulatory or cross-domain system of record.
Google Workspace collaboration is a primary requirement.	Complex reusable metrics or governed drill paths must be modeled.
The audience is bounded and the report has an accountable owner.	Restricted data cannot meet sharing, connector or perimeter controls.

Use when	Do not use when
Performance and concurrency have been tested for the selected connector.	The report relies on many extracts or duplicated calculations.

10.2 Connected Sheets

Connected Sheets enables approved users to analyze BigQuery data through spreadsheet features without manually exporting entire datasets. It is appropriate for bounded ad hoc analysis, reconciliation and familiar pivot-table workflows. Access remains governed by BigQuery IAM and data policies. For VPC Service Controls-protected data, implementation teams must validate required ingress and egress behavior because result handling interacts with Google Sheets.

Control area	Required control
Source	Only certified product views or explicitly approved reconciliation datasets.
Identity	Named workforce identity with BigQuery permissions; no shared credentials.
Classification	Public, Internal and approved Confidential by default; Restricted/Regulated require security and privacy approval.
Volume	Query and result limits; no use as bulk extraction mechanism.
Sharing	Workspace sharing restricted to approved organization/groups; external sharing disabled unless contracted.
Logic	Workbook formulas are not certified KPI definitions. Reusable logic must move to the semantic layer.
Lifecycle	Owner, purpose, review date and automatic cleanup for inactive workbooks.
Audit	BigQuery jobs and relevant Workspace sharing events retained according to policy.

VPC Service Controls dependency. Connected Sheets for protected BigQuery data requires explicit validation of access levels and ingress/egress policies. It is not assumed to work merely because a user can query BigQuery in another interface.

11. BigQuery Studio, SQL Workspaces and Notebooks

BigQuery Studio is the advanced self-service environment for analytics engineers, data scientists and power analysts who require SQL, notebooks and managed workspace assets. It complements Looker; it does not replace the certified semantic layer. Production datasets remain read-only for analysts except through controlled deployment identities.

Capability	Approved use	Guardrails
Saved SQL queries	Reusable analysis, validation and operational troubleshooting.	Repository or managed asset, owner, parameterization, dry run and maximum bytes.
SQL workspace	Interactive development against certified products.	Dedicated billing project, labels, regional location and no raw access by default.
Notebooks	Reproducible analysis and statistical exploration.	Approved runtime, dependency control, sensitive-data restrictions and output lifecycle.
Data canvas/preparation	Assisted exploration and transformation.	Nonproduction by default; inspect generated SQL and do not publish unreviewed logic.
Team folders	Shared analytical assets.	Group ownership, least privilege, naming and quarterly review.
Temporary outputs	Intermediate tables for analysis.	Sandbox dataset, expiration, classification and cost attribution.

11.1 SQL workspace standard

Standard	Requirement
Project	Query from an approved analytics consumer project, not the data-owner project unless justified.

Standard	Requirement
Location	Job location must match the dataset and approved geography.
Cost	Use labels, maximum bytes billed, partition filters and monitored budgets.
Security	No user-managed service-account keys; group access and short-lived credentials.
Data	Certified products first; raw/bronze requires engineering purpose and approval.
Output	Temporary tables expire; persistent output is created through product/CI/CD workflow.
Testing	Dry run, representative execution, row-count/control checks and peer review for reusable SQL.
Observability	Job metadata retained and attributable to user, product, purpose and channel.

11.2 Representative BigQuery SQL

```
-- Purpose: publish an analytics-ready semantic view with stable names.
CREATE OR REPLACE VIEW `acme-data-sales-prod-amer-01.sales_semantic.v_order_performance` AS
SELECT
  DATE(order_ts) AS order_date,
  region_code,
  channel_code,
  COUNT(DISTINCT order_id) AS order_count,
  SUM(net_revenue_amount) AS net_revenue_amount,
  SUM(margin_amount) AS margin_amount
FROM `acme-data-sales-prod-amer-01.sales_curated.order_fact`
WHERE is_current = TRUE
GROUP BY 1,2,3;

-- Acceptance test: no duplicate business grain.
ASSERT (SELECT COUNT(*) FROM (
  SELECT order_date, region_code, channel_code, COUNT(*) c
  FROM `acme-data-sales-prod-amer-01.sales_semantic.v_order_performance`
  GROUP BY 1,2,3 HAVING c > 1)) = 0
AS 'Semantic view contains duplicate business grain';
```

12. BI Engine, Caching, Concurrency and Query Performance

Performance is managed as an end-to-end property: source-product physical design, semantic-model correctness, query shape, acceleration eligibility, workload capacity and consumer concurrency. The platform does not purchase capacity to compensate for unbounded joins or unpartitioned scans.

12.1 Performance design sequence

Order	Optimization	Decision rule
1	Correct grain, keys and join relationships.	No acceleration proceeds until fanout and many-to-many behavior are understood.
2	Select only required columns and filters.	Certified content must demonstrate partition pruning and avoid SELECT *.
3	Partition, cluster and pre-aggregate product interfaces.	Design for dominant access paths while preserving product contract.
4	Use materialized views or aggregate tables.	Use for stable, repeated aggregation patterns with measurable benefit and controlled freshness.
5	Use BigQuery query cache.	Treat as opportunistic; do not base SLOs solely on cached results.
6	Use BI Engine for eligible, repeated BI access.	Size after measurement; define preferred tables where appropriate; test security compatibility.
7	Use reservations, autoscaling and assignments.	Isolate critical workloads and set capacity based on observed slot and queue metrics.

Order	Optimization	Decision rule
8	Tune Looker persistence and caching.	Align datagroups/persistence with product freshness and change semantics.

12.2 BI Engine

BI Engine provides in-memory acceleration for eligible BigQuery BI workloads. It is a conditional optimization, not a separate data store or governance boundary. Capacity is sized from observed working sets and concurrency. Preferred tables may be configured for priority assets, but implementation teams must confirm current feature eligibility. Row-level security or unsupported query shapes can prevent acceleration, while BigQuery masking remains enforced. Therefore, performance tests must use the real security model rather than administrator accounts only.

Design concern	Guidance
Selection	Use for repeated, latency-sensitive dashboards with eligible SQL and stable working sets.
Sizing	Measure table working set, hot columns, concurrent queries, eviction and acceleration statistics.
Freshness	Confirm that acceleration behavior meets source-product freshness and report SLO.
Security	Test each persona with RLS, policy tags and masking; never assume administrator performance generalizes.
Cost	Track reserved BI Engine capacity, utilization and latency improvement by product.
Failure	Dashboard remains functionally correct without acceleration, though it may breach performance SLO; alert and fail over to standard BigQuery execution.

12.3 Query caching and Looker persistence

BigQuery cached results are enabled by default for eligible identical deterministic queries when referenced tables have not changed. Because cache reuse is conditional, it is treated as a cost and latency optimization rather than a reliability guarantee. Looker query cache, persisted derived tables and datagroups must be aligned to data-product update signals. Stale-while-unaware is prohibited for regulatory and critical operational reporting.

Cache type	Use	Control
BigQuery query cache	Repeated identical eligible queries.	Monitor cache hit behavior; do not use for freshness-sensitive validation.
Looker query cache	Repeated semantic queries and dashboard use.	Datagroup tied to product freshness; explicit max age and invalidation.
Persisted derived table	Complex reusable transformation when native product/aggregate is not appropriate.	Owner, trigger, build identity, failure monitoring, cost and rollback.
Materialized view	Native incremental precomputation for stable patterns.	Smart-tuning eligibility, refresh, partition alignment, security and cost review.
BI Engine	In-memory acceleration.	Capacity, preferred tables, security compatibility and usage telemetry.

12.4 Concurrency and reservations

BigQuery reservations provide a capacity and workload-management boundary. The baseline uses separate assignments for production BI, ad hoc analysis and data engineering when contention risk or predictable performance justifies it. Baseline slots and autoscaling limits are derived from load tests and actual slot consumption. Target concurrency, where configured, is a workload-shaping input rather than an exact hard guarantee and must be tested.

Workload class	Priority	Capacity pattern	Example acceptance criterion
Tier 0 executive/regulatory	Highest	Dedicated reservation or protected assignment with headroom.	p95 dashboard under 5 seconds at 2x forecast peak; no queue breach.
Tier 1 operational BI	High	Protected production BI reservation, autoscaling bounded.	p95 under 8 seconds; freshness and queue SLO maintained.
Domain governed exploration	Normal	Shared domain/analytics reservation.	p95 Explore under 15 seconds for standard queries.
Ad hoc / notebook	Lower	Separate consumer project and reservation or on-demand policy.	Cannot exhaust Tier 0/1 capacity; maximum byte controls active.

Workload class	Priority	Capacity pattern	Example acceptance criterion
Engineering/backfill	Batch	Separate reservation and schedule.	No material impact on interactive BI; backfill window met.

12.5 Query-performance standards

ID	Mandatory standard	Evidence
PERF-01	Certified queries use partition filters when partitioned tables are accessed.	Dry-run bytes and query plan.
PERF-02	No unbounded many-to-many joins in certified Explores.	Model relationship test.
PERF-03	Critical dashboards are tested with production-scale data and real persona security.	Load-test report.
PERF-04	Query scans, slot time and queue time have product thresholds.	Monitoring configuration.
PERF-05	Cross-region data access is prohibited unless explicitly designed and approved.	Architecture decision and data-flow evidence.
PERF-06	Large repeated aggregations use an approved aggregate strategy where cost-beneficial.	Before/after benchmark and ownership.
PERF-07	Ad hoc environments apply maximum bytes and temporary-table expiry.	Project/job configuration and audit query.
PERF-08	Capacity changes include forecast, unit economics, rollback and review date.	FinOps approval record.

12.6 Terraform capacity example

```
terraform {
  required_version = ">= 1.6.0"

  required_providers {
    google = {
      source  = "hashicorp/google"
      version = "~> 6.0"
    }
  }
}

provider "google" {
  project = var.project_id
  region  = var.region
}

resource "google_bigquery_reservation" "bi" {
  name              = "bqrs-analytics-prod-amer"
  location          = var.bigquery_location
  edition           = "ENTERPRISE"
  baseline_slots    = var.baseline_slots
  ignore_idle_slots = false
  concurrency       = var.target_concurrency

  autoscale {
    max_slots = var.max_slots
  }
}

resource "google_bigquery_reservation_assignment" "analytics_project" {
  assignee    = "projects/${var.analytics_project_number}"
  job_type    = "QUERY"
  reservation = google_bigquery_reservation.bi.id
}

resource "google_bigquery_bi_reservation" "accelerator" {
  location = var.bigquery_location
  size     = var.bi_engine_bytes

  preferred_tables {
    project_id = var.data_project_id
    dataset_id = "sales_semantic"
    table_id   = "order_performance_daily"
  }
}
```

13. Dashboard, Report and Use-Case Standards

Analytics assets are engineered according to decision risk. A dashboard is not certified because it is visually polished; it is certified because its measures, data, security, performance, ownership and operational behavior are controlled.

13.1 Dashboard standards

Area	Standard
Purpose	The dashboard states the decision, audience, owner and support tier.
Metrics	Tier 0/1 dashboards use certified measures; local calculations are documented and bounded.
Context	Shows as-of time, data freshness, filters, timezone, currency and material caveats.
Layout	Critical KPIs and exceptions first; consistent visual grammar; avoid decorative density.
Drill	Drill paths preserve security, grain and explanatory context.
Performance	Tile count, query fan-out and refresh behavior meet tested SLOs.
Accessibility	Readable contrast, labels, keyboard/navigation considerations and alternative text where supported.
Sharing	Folder, schedule, export and external sharing follow classification.
Lifecycle	Version, owner, review date, usage and deprecation state are recorded.
Failure	Displays stale-data or unavailable state; operational alternatives documented for critical decisions.

13.2 Executive dashboards

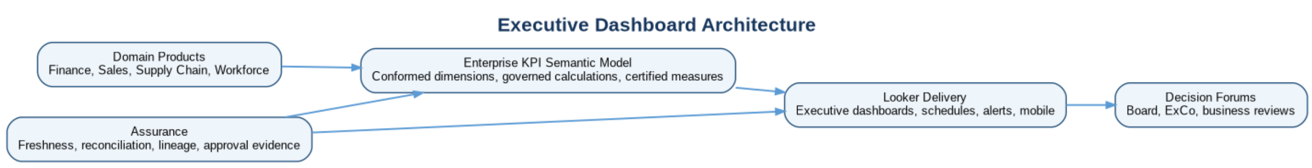

Figure 4 - Executive dashboard architecture and assurance flow

- Use a small set of outcome KPIs aligned to business objectives and accountable owners.
- Provide variance, trend, target, explanatory dimensions and linked action paths.
- Freeze definitions during board or regulatory reporting windows through controlled release management.
- Show timestamp, reporting basis, scope and material adjustments.
- Maintain an executive issue path for disputed figures; never silently overwrite historical reports.

13.3 Operational analytics

Operational dashboards support near-real-time decisions and exception handling. They require freshness and availability SLOs, bounded query patterns, incident routing and an alternative operational view when analytics is impaired. They must not become a substitute for a transactional system when sub-second response, write-back or hard real-time control is required.

13.4 Regulatory reporting

Control	Requirement
Definition control	Measures, reporting basis and regulatory interpretation are approved and versioned.

Control	Requirement
Data lineage	Source-to-report lineage and transformations are retained.
Reconciliation	Control totals, exceptions, adjustments and sign-off evidence are preserved.
Access	Least privilege, segregation of duties and controlled report release.
Reproducibility	Report version, parameters, data cutoff and code version can be reconstructed.
Retention	Outputs and evidence retained under records-management and legal-hold rules.
Change freeze	Defined freeze window and emergency change process.
Export	Approved secure format, delivery channel, recipient and audit trail.

13.5 Embedded analytics

Embedded analytics uses Looker APIs and supported embedding patterns to deliver governed analytics inside enterprise or customer applications. Signed embed URLs or cookieless embedding are generated server-side and tied to application authentication and entitlements. A signed URL is treated as a sensitive, short-lived artifact; it is never generated in browser code or logged. Tenant isolation is enforced using authoritative data controls and validated with negative tests.

Concern	Required control
Authentication	Application authenticates the user; backend maps identity to approved Looker embed attributes.
Authorization	Tenant, role, geography and product entitlements applied in both semantic and data layers.
Token/URL	Generated server-side, short-lived, replay-resistant according to product semantics and excluded from logs.
Origin	Approved embed domains and content security policy.
Load	Per-tenant and aggregate concurrency/load tests; protection against abusive query patterns.
Observability	Correlate application session, Looker query and BigQuery job without exposing sensitive tokens.
Support	Application and analytics teams define incident ownership, degraded mode and customer communications.

14. Analytics Security and Access

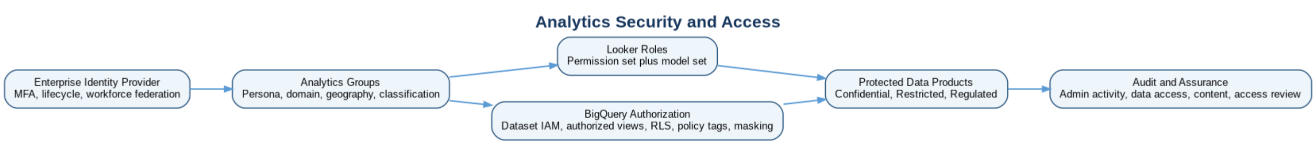

Figure 5 - Layered analytics identity, authorization and audit architecture

Analytics security uses verified identity, group-based authorization, least privilege, regional boundaries and defense in depth. Looker content permissions determine who can see and edit analytics assets; Looker roles and model sets determine platform capabilities and model visibility; BigQuery IAM and data policies determine authoritative access to data.

Layer	Control	Mandatory rule
Identity	Enterprise IdP, MFA, lifecycle and federation.	No unmanaged local production users except tested break-glass identities.
Group	Persona + domain + geography + classification groups.	No routine individual production grants.
Looker role	Permission set and model set.	Separate administration, development, scheduling and consumption.
Content	Folder and content permissions.	Certified folders are group-owned and protected from direct editing.
BigQuery	Dataset/table/view IAM and job permissions.	Connections and users receive minimum required access.
Row	Row-level access policies or authorized interfaces.	Enforce tenant, legal entity, geography or purpose where required.
Column	Policy tags, column-level access and masking.	Protect sensitive attributes independent of visualization.
Perimeter/network	VPC Service Controls and private connectivity when applicable.	Validate supported services and narrow ingress/egress; do not use broad bridges.

Layer	Control	Mandatory rule
Audit	Looker activity, BigQuery jobs, Data Access logs and access reviews.	Retain and route according to classification and compliance.

14.1 Analytics access matrix

Role	Looker permissions	Models/data	Content rights	High-risk actions
Viewer	View dashboards, run approved queries.	Assigned model set; data policies apply.	View certified folders.	No download of restricted data by default.
Explorer	Explore and save personal/team content.	Certified domain models.	Create in delegated folders.	Schedules/exports bounded.
Content curator	Publish team/domain content.	Approved models.	Manage delegated domain folder.	Cannot administer connections or roles.
Analytics developer	Develop LookML in branches.	Development connection and approved validation access.	Development folders.	No direct production deploy.
Release manager	Promote approved release.	Production models through deployment workflow.	Certified folder release.	Separated from primary author where required.
Platform admin	Manage instance configuration.	Administrative scope only as needed.	System administration.	Time-bound privileged access and enhanced audit.
Auditor	Read evidence and system activity.	No business data unless separately approved.	Read controlled reports/evidence.	No modification.

14.2 Classification-based channel policy

Classification	Looker	Data Studio Pro	Connected Sheets	Export/embed
Public	Allowed with publishing approval.	Allowed.	Allowed.	External release requires owner approval.
Internal	Allowed.	Allowed in managed workspace.	Allowed in managed workspace.	External prohibited by default.
Confidential	Allowed with group control and audit.	Conditional.	Conditional and limited.	Approved secure export only.
Restricted	Allowed in approved model/instance with layered controls.	Exception-based.	Strong exception; normally prohibited.	No external sharing; embed only after security review.
Regulated	Controlled instance/model and report process.	Normally not baseline.	Normally prohibited.	Regulator/contract-specific controlled delivery.

14.3 Security testing

- Positive and negative entitlement tests for every persona and geography.
- Direct BigQuery tests proving that hidden BI fields remain protected.
- RLS and masking tests using realistic users, not only administrators.
- External sharing, schedule, download and embed tests by classification.
- VPC Service Controls dry-run analysis and explicit Connected Sheets/third-party path validation.
- Privileged access, break-glass and access-review evidence.
- Injection, malformed filter, URL/token handling and tenant-isolation tests for embedded analytics.

15. Analytics Governance, Testing and Deployment

Analytics governance combines business ownership, semantic standards, technical controls and content lifecycle management. It is federated: enterprise policy and shared KPI standards are central, while domains implement and operate product-aligned models within those guardrails.

Role	Accountability
Enterprise Analytics Council	Channel strategy, semantic standards, cross-domain conflicts and investment priorities.
KPI owner	Definition, decision use, approval scope, target and change decision.

Role	Accountability
Data-product owner	Source interface, contract, quality, freshness, security and lifecycle.
Domain analytics lead	Model roadmap, delivery quality, support, content hygiene and cost.
Analytics engineer	LookML/SQL implementation, tests, documentation and performance.
Platform team	Looker, capacity, integration, templates, monitoring and paved roads.
Data governance	Glossary, certification policy, lineage, stewardship and exceptions.
Security/privacy	Classification, access design, embedding, export and compliance assurance.
SRE/operations	SLOs, alerting, incidents, runbooks and capacity coordination.
FinOps	Allocation, forecast, anomaly detection, unit economics and optimization.

15.1 Testing pyramid

Test layer	Examples	Release gate
Static / lint	LookML syntax, naming, descriptions, forbidden constructs, repository policy.	All commits.
Model validation	LookML validator, SQL validation, primary key and join relationship checks.	Pull request.
Data tests	Uniqueness, non-null, accepted values, referential integrity and KPI assertions.	Pull request and scheduled production.
Reconciliation	Certified KPI versus independent control or prior approved report.	Certification and material change.
Content validation	Broken fields, deleted content, impacted dashboards and schedules.	Release candidate.
Security	Persona access, RLS, masking, downloads, sharing and embed tenant isolation.	Release candidate; recurring control test.
Performance	Dashboard p95, query bytes, slot time, queue time and concurrency.	Before Tier 0/1 release and major scale change.
Resilience	Connection failure, capacity constraint, stale data, rollback and recovery.	Production readiness and annual exercise.
User acceptance	Decision owner validates meaning, workflow and usability.	Business approval.

15.2 Deployment workflow

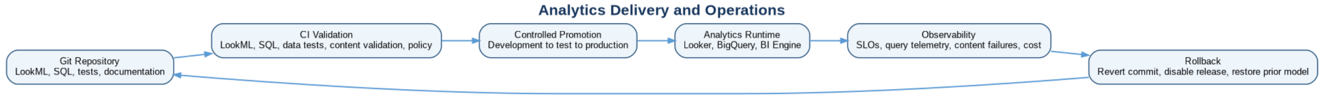

Figure 6 - Analytics semantic-model delivery and operational feedback loop

Stage	Activities	Exit evidence
Develop	Feature branch, development connection, unit/model tests and documentation.	Local/branch validation passes.
Pull request	Peer review, code-owner approval, automated validators, security and policy checks.	Approved PR and immutable build evidence.
Integration	Compile against test data, content validation, reconciliation and regression.	Integration report with no unresolved critical issue.
Performance/security	Representative personas, production-scale data, concurrency and negative access tests.	Signed test evidence.
Release approval	Product owner, KPI owner and control approver according to tier.	Change/release record.
Promote	Deploy tagged release; update catalog/version; run smoke tests.	Production verification.
Observe	Monitor SLO, errors, BigQuery jobs, cost and user impact.	No release-triggered breach.
Rollback	Revert model/content release or disable affected feature; communicate.	Prior stable version restored and incident recorded.

15.3 Analytics deployment standards

- Production changes are promoted through an approved deployment mechanism; console-only semantic changes are noncompliant.
- Connection, instance, reservation and IAM changes use Terraform or approved API automation where supported.
- The same source commit produces test and production artifacts; environment-specific configuration is externalized.
- Production service identities are not used in developer workstations.
- Content promotion preserves stable identifiers where the platform and release process support them.
- Rollback is tested before Tier 0/1 release and includes semantic, content, capacity and access changes.

16. Analytics SLOs, Observability and Support

Analytics reliability is measured from the consumer's perspective and decomposed into data freshness, semantic correctness, query success, response time, content delivery and support. Google Cloud service availability does not by itself prove that an executive dashboard is reliable.

Tier	Example workload	Availability target	Performance target	Freshness / accuracy
Tier 0	Board, regulatory, critical operational decisions.	99.95% during defined decision window.	Dashboard p95 <= 5 s; critical page <= 8 s.	Product-specific freshness; 100% control reconciliation within approved tolerance.
Tier 1	Enterprise and domain management dashboards.	99.9% monthly.	Dashboard p95 <= 8 s; Explore p95 <= 15 s for standard query.	Freshness SLO from product contract; quality threshold >= approved score.
Tier 2	Team reports and ad hoc analysis.	99.5% or business-hours objective.	Best effort with query safety threshold.	Source freshness displayed; no certified accuracy claim unless validated.

Architecture targets, not provider SLA. These are enterprise workload objectives that must be validated against selected Looker and Google Cloud editions, product SLAs, regional deployment, support plan and measured capacity.

16.1 SLI catalog

SLI	Measurement	Owner	Alert condition
Dashboard availability	Successful dashboard loads / valid attempts.	Analytics product owner.	Error-budget burn or decision-window failure.
Query success	Successful Looker/BigQuery queries / total.	Platform and domain team.	Tier threshold breached.
End-to-end latency	User request to rendered critical content.	Platform/SRE.	p95/p99 breach.
Freshness	Current time minus certified product watermark.	Data-product owner.	Contract threshold exceeded.
Accuracy/reconciliation	Difference versus approved control total.	KPI owner/controller.	Tolerance breached.
Queue time	BigQuery pending/queue duration.	Platform capacity owner.	Sustained threshold.
Acceleration	BI Engine/cache/aggregate effectiveness.	Performance engineering.	Regression with SLO or cost impact.
Schedule delivery	Successful scheduled report deliveries.	Content owner.	Critical delivery failure.
Cost efficiency	Bytes/slot time/cost per dashboard or active user.	FinOps and product owner.	Budget anomaly or unit-cost regression.
Content health	Broken content, orphaned assets and inactive schedules.	Domain analytics lead.	Critical content break or hygiene backlog.

16.2 Monitoring and alerting

Severity	Example	Routing	Response
SEV-1	Tier 0 decision window unavailable or materially incorrect regulated report.	24x7 incident commander, analytics, data product, security/compliance as needed.	Immediate containment, executive communication and recovery.
SEV-2	Tier 1 widespread failure, severe latency or missed critical schedule.	On-call analytics/platform and product owner.	Restore within tier target; assess error budget.
SEV-3	Single report failure, degraded performance or noncritical stale data.	Domain support queue.	Business-hours remediation and trend review.

Severity	Example	Routing	Response
SEV-4	Content hygiene, documentation, optimization or low-risk defect.	Backlog owner.	Planned release.

16.3 Support model

Level	Responsibility
L0 Self-service	Catalog, dashboard help, definitions, known issues and status communication.
L1 Analytics service desk	Triage access, content and user issues; route by product metadata.
L2 Domain analytics	Semantic model, dashboard, KPI and product-specific support.
L3 Platform engineering/SRE	Looker instance, connections, capacity, APIs, deployment and systemic performance.
L4 Vendor/support	Product defects, quota/support cases and provider escalation.

17. Google Cloud Service Assessments

The following assessments apply the 32-field structure required by the master prompt. Quotas, feature support, availability, licensing and pricing are deliberately not hard-coded where they can change; implementation teams must validate the current official documentation and commercial agreement.

17.1 Looker

Governed semantic modeling, exploration, BI delivery and embedded analytics.

Assessment field	Guidance
Service name	Looker
Purpose	Governed semantic modeling, exploration, BI delivery and embedded analytics.
Enterprise architecture role	Primary enterprise semantic and governed BI platform.
Business capabilities supported	Executive dashboards, operational analytics, regulatory reporting, governed exploration, alerts and embedded analytics.
Technical capabilities	LookML, Explores, dashboards, schedules, APIs, Git workflows, permissions, model sets and embedding.
Benefits	Reusable business logic, code-based semantics, strong BigQuery integration and broad delivery patterns.
Limitations	Licensing/edition considerations, semantic-model skills, content sprawl risk and dependence on correct underlying data design.
Service quotas	Validate API, schedule, query, embed and instance limits for the selected edition.
Regional availability considerations	Select instance/connection topology according to residency, latency and supported regions.
High-availability characteristics	Managed service characteristics depend on edition/topology; workload SLO must include data and connection dependencies.
Disaster-recovery considerations	Export/version LookML in Git, document configuration, protect keys/secrets, back up critical content where supported and test reconstruction.
Security model	Roles combine permissions and model access; content permissions add folder control; BigQuery remains authoritative.
IAM requirements	Federated users/groups, least-privilege admin/developer/viewer roles and controlled connection identity.
Network requirements	Approved connectivity to BigQuery and enterprise identity; private connectivity where supported.
VPC Service Controls considerations	Validate Looker/BigQuery access and perimeter ingress/egress for the chosen deployment.
Encryption capabilities	Encryption in transit and at rest per product; validate CMEK and edition support.
CMEK support	Conditional; verify target deployment and dependency support.
Logging	System activity, admin/content events and correlated BigQuery audit logs.
Monitoring	Instance health, query failures, latency, content errors, schedules, API and adoption.
Alerting	SLO burn, connection failure, schedule failure, query-error spikes and capacity impact.
Cost drivers	Licensed users/platform, BigQuery queries, BI capacity, schedules, embeds and operational support.
Cost optimization techniques	Model efficiency, content hygiene, cache strategy, user/license management and capacity attribution.

Assessment field	Guidance
Operational responsibilities	Platform team operates instance; domains operate models/content; security governs access; SRE owns systemic reliability.
Integration points	BigQuery, Git, identity provider, catalog, CI/CD, monitoring, applications and Workspace.
Alternatives	Data Studio Pro or third-party BI for bounded requirements; direct SQL/notebooks for technical analysis.
Selection criteria	Choose when governed reusable semantics, exploration, delivery or embedding are material.
Best practices	Product-aligned models, Git, tests, group roles, certified folders, real-user performance tests and lifecycle management.
Reference architecture	Certified BigQuery products -> LookML model -> governed content/embed -> telemetry and FinOps.
Sample configuration	See examples/looker.
Common pitfalls	Monolithic model, dashboard calculations as KPIs, hidden-field security, individual ownership and unmanaged schedules.
Anti-patterns	Direct raw access, shared credentials, production edits without review and unrestricted downloads.
Production-readiness checklist	Edition/region approved; access and connection tested; CI/CD active; SLO/runbook/DR/cost ownership established.

17.2 BigQuery

Governed analytical storage, SQL execution and workload platform.

Assessment field	Guidance
Service name	BigQuery
Purpose	Governed analytical storage, SQL execution and workload platform.
Enterprise architecture role	Authoritative analytical data plane for certified products and semantic interfaces.
Business capabilities supported	Interactive analytics, executive/reporting queries, ad hoc SQL, data sharing and analytics acceleration.
Technical capabilities	Native tables/views, materialized views, reservations, BI Engine, RLS, policy tags, masking and INFORMATION_SCHEMA.
Benefits	Petabyte-scale serverless analytics, workload controls, integrated governance/security and SQL ecosystem.
Limitations	Query cost/performance depend on design; location boundaries; unsuitable for transactional serving patterns.
Service quotas	Validate query, API, reservation, table, job and concurrent operation quotas.
Regional availability considerations	Datasets and jobs are location-bound; align with AMER/EMEA/APAC residency and avoid unplanned cross-region movement.
High-availability characteristics	Managed regional/multi-region service characteristics; design workloads to tolerate dependency and region failures.
Disaster-recovery considerations	Product tier determines replication/export/snapshot/rebuild; semantic code and IaC support reconstruction.
Security model	Project/dataset/table/view IAM, RLS, policy tags, data policies, masking, authorized views and audit.
IAM requirements	Separate job-user, data-viewer, data-policy and administration duties; group/service identities.
Network requirements	Private Google API access and approved service perimeter paths for restricted workloads.
VPC Service Controls considerations	Place sensitive projects/services in validated perimeters with narrow ingress/egress.
Encryption capabilities	Default encryption; CMEK where supported and inherited requirement justifies it.
CMEK support	Supported for relevant resources with location/key and operational dependencies to validate.
Logging	Admin Activity and Data Access audit logs, jobs and reservation telemetry.
Monitoring	Query failures, bytes, slots, queue, latency, storage, freshness and cost.
Alerting	SLO, cost anomaly, queue saturation, failed jobs, quota and security events.
Cost drivers	Query compute or capacity, storage, BI Engine, data transfer, streaming and extracts.
Cost optimization techniques	Partitioning, clustering, selective SQL, materialized views, reservations, cache, lifecycle and workload attribution.
Operational responsibilities	Data product teams own tables/interfaces; platform owns shared capacity/standards; FinOps reviews usage.
Integration points	Looker, Data Studio Pro, Connected Sheets, BigQuery Studio, Dataform, catalog and Vertex AI.
Alternatives	Other warehouses or Spark only when documented sovereignty, feature or economic requirement exists.
Selection criteria	Default for governed analytical products and SQL serving.
Best practices	Separate owner/consumer projects, stable semantic interfaces, tests, labels, query safety and capacity testing.
Reference architecture	Domain datasets -> semantic views/tables -> reservations/BI Engine -> analytics channels.

Assessment field	Guidance
Sample configuration	See examples/sql and examples/terraform.
Common pitfalls	SELECT *, unpartitioned scans, cross-region queries, one shared project and direct user grants.
Anti-patterns	Using BigQuery as an OLTP system or granting broad access to raw data for convenience.
Production-readiness checklist	Location, security, capacity, quota, cost, recovery, monitoring and load tests approved.

17.3 BI Engine

In-memory acceleration for eligible BigQuery BI workloads.

Assessment field	Guidance
Service name	BI Engine
Purpose	In-memory acceleration for eligible BigQuery BI workloads.
Enterprise architecture role	Conditional acceleration layer for high-value repeated dashboard queries.
Business capabilities supported	Low-latency executive and operational dashboards at concurrency.
Technical capabilities	Capacity reservation, preferred tables and BigQuery-integrated acceleration.
Benefits	Can reduce latency and repeated compute for eligible workloads.
Limitations	Eligibility varies by SQL/features; not all security/query patterns accelerate; requires capacity management.
Service quotas	Validate capacity and regional limits.
Regional availability considerations	Reservation location must align with BigQuery datasets/workloads.
High-availability characteristics	Managed acceleration; queries should remain correct through standard BigQuery when acceleration is unavailable.
Disaster-recovery considerations	Recreate capacity through Terraform; dashboards degrade gracefully to BigQuery where possible.
Security model	BigQuery security remains enforced; test real users with RLS/masking.
IAM requirements	Capacity administration separated from query consumption.
Network requirements	No separate user network path beyond BigQuery integration.
VPC Service Controls considerations	Inherited through BigQuery; verify current support.
Encryption capabilities	Inherited from BigQuery service behavior.
CMEK support	Validate current dependency behavior.
Logging	BigQuery job and acceleration statistics.
Monitoring	Utilization, acceleration ratio, latency, eviction and fallback.
Alerting	Capacity saturation or loss of acceleration causing SLO breach.
Cost drivers	Reserved memory capacity by location.
Cost optimization techniques	Right-size working set, preferred tables, remove unused capacity and compare latency benefit.
Operational responsibilities	Platform/performance team owns capacity; domain teams own query/model efficiency.
Integration points	BigQuery, Looker and Data Studio connectors.
Alternatives	Materialized views, aggregate tables, query cache and more BigQuery slots.
Selection criteria	Repeated eligible BI queries with measurable latency/business value.
Best practices	Benchmark before/after, test security personas, monitor eligibility and retain fallback.
Reference architecture	Hot semantic tables -> BI Engine -> dashboard channel; standard BigQuery fallback.
Sample configuration	See Terraform example.
Common pitfalls	Sizing by total table size, testing only admin users and expecting every query to accelerate.
Anti-patterns	Buying BI Engine before correcting query/model design.
Production-readiness checklist	Eligibility, capacity, cost, security and fallback tests pass.

17.4 Data Studio Pro

Managed lightweight visualization and collaboration channel.

Assessment field	Guidance
Service name	Data Studio Pro
Purpose	Managed lightweight visualization and collaboration channel.
Enterprise architecture role	Conditional secondary BI channel on certified sources.
Business capabilities supported	Departmental reporting, lightweight visualization and Workspace collaboration.
Technical capabilities	Managed reports, data sources, BigQuery connectivity, team workspaces and administration features according to edition.
Benefits	Accessible authoring and collaboration for simple use cases.
Limitations	Not the primary code-governed semantic authority; connector and governance limits require validation.
Service quotas	Validate connector, extract, refresh and report limits.

Assessment field	Guidance
Regional availability considerations	Confirm data processing and connector behavior for residency-sensitive workloads.
High-availability characteristics	Managed SaaS; workload continuity depends on source and connector behavior.
Disaster-recovery considerations	Retain report inventory, ownership and source definitions; define manual/rebuild process for critical assets.
Security model	Workspace identity, data-source credentials, sharing controls and BigQuery authorization.
IAM requirements	Managed organization/group access and least-privilege BigQuery source access.
Network requirements	Approved service access to BigQuery; validate protected-data connectivity.
VPC Service Controls considerations	Validate connector support and perimeter path before sensitive use.
Encryption capabilities	Managed service encryption; validate contractual requirements.
CMEK support	Validate current product support; do not assume.
Logging	Workspace/admin sharing events where available plus BigQuery query logs.
Monitoring	Report failures, source refresh, usage and query cost.
Alerting	Source failure, stale report and cost anomaly through available integrations.
Cost drivers	Licensing, BigQuery queries, extracts and support.
Cost optimization techniques	Certified sources, limit extracts, report hygiene and query-efficient views.
Operational responsibilities	Platform governs service; departments own reports; data teams own sources.
Integration points	BigQuery, Workspace, Looker semantic sources where supported and identity.
Alternatives	Looker for governed enterprise BI; Connected Sheets for spreadsheet analysis.
Selection criteria	Lightweight bounded use with no need for new authoritative metrics.
Best practices	Managed team content, certified sources, sharing policy and lifecycle.
Reference architecture	Certified source -> managed data source -> team report -> audit/usage.
Sample configuration	Use access and KPI matrices in package.
Common pitfalls	Ownerless reports, embedded calculations, broad sharing and extracts.
Anti-patterns	Using it as the regulatory reporting or enterprise metric system of record.
Production-readiness checklist	Edition, sharing, source, performance, support and classification approved.

17.5 Connected Sheets and BigQuery Studio

Controlled spreadsheet analysis and advanced SQL/notebook workspaces.

Assessment field	Guidance
Service name	Connected Sheets and BigQuery Studio
Purpose	Controlled spreadsheet analysis and advanced SQL/notebook workspaces.
Enterprise architecture role	Persona-specific self-service channels consuming certified BigQuery products.
Business capabilities supported	Reconciliation, pivot analysis, advanced SQL, notebooks and exploratory analysis.
Technical capabilities	Live BigQuery-connected sheets, saved queries, notebooks and team workspace assets.
Benefits	Familiar analyst experiences without requiring bulk extracts for every question.
Limitations	Spreadsheet logic is difficult to govern; notebook/runtime and connector controls vary; not a semantic authority.
Service quotas	Validate Sheets, connector, BigQuery and notebook/runtime limits.
Regional availability considerations	Jobs and outputs must remain in approved BigQuery location; Workspace behavior requires assessment.
High-availability characteristics	Depends on Workspace/BigQuery and notebook/runtime components; not all exploratory assets receive production SLO.
Disaster-recovery considerations	Reusable SQL/notebooks in managed repository/assets; workbooks have ownership and retention.
Security model	Workforce identity plus BigQuery IAM/data policies; Workspace sharing adds another control surface.
IAM requirements	Named users/groups, approved consumer projects and least privilege.
Network requirements	Approved Google service paths; notebook runtimes follow landing-zone controls.
VPC Service Controls considerations	Connected Sheets requires explicit ingress/egress validation for protected BigQuery data.
Encryption capabilities	Inherited from BigQuery/Workspace/runtime; validate classification requirements.
CMEK support	Varies by component; validate current support.
Logging	BigQuery jobs, Workspace sharing/admin logs and runtime logs where applicable.
Monitoring	Queries, bytes, failures, inactive assets, sharing and runtime health.
Alerting	Cost anomaly, policy violation, stale critical workbook or notebook failure.
Cost drivers	BigQuery compute, notebook runtime, storage, extracts and operational support.
Cost optimization techniques	Byte limits, partition filters, temporary expiry, live connections and cleanup.
Operational responsibilities	Users own exploratory assets; platform owns workspace standards; domain teams own source products.
Integration points	BigQuery, Workspace, Git/repositories, catalog and Vertex AI in Volume 7.

Assessment field	Guidance
Alternatives	Looker for reusable governed content; Data Studio Pro for lightweight visualization.
Selection criteria	Use when persona workflow genuinely requires spreadsheet or code-centric analysis.
Best practices	Certified sources, owner/expiry, restricted sharing, query safety and reproducibility.
Reference architecture	User -> approved workspace/consumer project -> certified BigQuery product -> telemetry.
Sample configuration	See SQL examples and access matrix.
Common pitfalls	Permanent workbook copies, unbounded queries, unmanaged notebooks and persistent personal tables.
Anti-patterns	Publishing spreadsheet/notebook calculations as enterprise KPIs without certification.
Production-readiness checklist	Access, location, cost, sharing, lifecycle and perimeter validation complete.

18. Architecture Decision Records

The following ADRs preserve inherited decisions and document analytics-specific choices. Dates and review triggers are explicit so implementation teams can reassess decisions after material product, scale, regulatory or commercial change.

ADR-ANA-001 - Adopt Looker and LookML as the primary enterprise semantic layer

Field	Decision record
ADR ID	ADR-ANA-001
Decision title	Adopt Looker and LookML as the primary enterprise semantic layer
Status	Approved
Date	11 July 2026
Architecture domain	Analytics / semantic architecture
Problem	The enterprise requires consistent KPIs, governed exploration and reusable semantics across thousands of analytical users without recreating logic in every dashboard.
Context	Global Fortune 100 analytics on inherited regional BigQuery data products and federated domain ownership.
Decision drivers	Trusted cross-domain measures; Code-based lifecycle and auditability; BigQuery integration; Federated domain ownership; Embedded and managed delivery
Constraints	Data residency, regulated workloads, thousands of users, existing BI coexistence and current product/licensing validation.
Assumptions	Certified data products exist; enterprise identity/groups and CI/CD foundations are available.
Alternatives considered	Looker/LookML; Semantic logic only in BigQuery views; Third-party BI semantic layer; Decentralized dashboard calculations
Evaluation criteria	Security, trust, scalability, usability, operability, cost, interoperability and reversibility.
Decision	Use Looker/LookML for certified enterprise and domain semantic models. Keep physical business rules that belong to the data product in BigQuery/Dataform. Data-layer security remains authoritative.
Rationale	Looker provides a governed reusable semantic model and delivery ecosystem while preserving BigQuery as the analytical data plane. It supports the inherited product-oriented and federated operating model.
Benefits	consistent definitions, reuse, governed exploration and auditable deployment.
Tradeoffs	licensing, specialist skills, model-governance effort and platform dependency.
Positive consequences	Positive: consistent definitions, reuse, governed exploration and auditable deployment.
Negative consequences	Negative: licensing, specialist skills, model-governance effort and platform dependency.
Risks	Poor model design may create fanout or slow queries.; Content can sprawl and metrics can still be duplicated outside certified models.
Mitigations	Semantic standards, automated tests, code owners, KPI registry, content lifecycle and performance engineering.
Implementation implications	Looker edition/topology and CI/CD must be funded; domain analytics teams require LookML capability.
Cost implications	Licensed platform plus BigQuery query/capacity cost; optimize models, users and content.
Security implications	Roles/model sets are additive to BigQuery controls; restricted data requires regional and perimeter validation.
Operational implications	Platform team runs Looker; domains own models/content; KPI owners approve measures.

Field	Decision record
Review triggers	Material Looker licensing/product change, new sovereignty requirement, unacceptable economics or inability to meet SLOs.
Review date	Quarterly, next formal review October 2026

ADR-ANA-002 - Use tiered analytics channels rather than a single-tool mandate

Field	Decision record
ADR ID	ADR-ANA-002
Decision title	Use tiered analytics channels rather than a single-tool mandate
Status	Approved
Date	11 July 2026
Architecture domain	Analytics consumption
Problem	Executives, business analysts, spreadsheet users, developers and embedded consumers have different needs. A single tool either over-constrains simple use cases or weakens controls for critical use cases.
Context	Global Fortune 100 analytics on inherited regional BigQuery data products and federated domain ownership.
Decision drivers	Persona fit; Governance; Cost; Collaboration; Advanced SQL/notebook requirements; Embedding
Constraints	Data residency, regulated workloads, thousands of users, existing BI coexistence and current product/licensing validation.
Assumptions	Certified data products exist; enterprise identity/groups and CI/CD foundations are available.
Alternatives considered	Looker only; Uncontrolled tool choice; Tiered portfolio: Looker, Data Studio Pro, Connected Sheets and BigQuery Studio
Evaluation criteria	Security, trust, scalability, usability, operability, cost, interoperability and reversibility.
Decision	Adopt a controlled portfolio. Looker is primary for certified enterprise analytics. Data Studio Pro, Connected Sheets and BigQuery Studio are conditional channels consuming certified sources.
Rationale	The portfolio supports legitimate workflows while keeping one semantic authority and a common security/cost model.
Benefits	better adoption and lower friction.
Tradeoffs	multiple control surfaces, skills and support paths.
Positive consequences	Positive: better adoption and lower friction.
Negative consequences	Negative: multiple control surfaces, skills and support paths.
Risks	Secondary tools may become shadow semantic layers.; Exports/workbooks can create uncontrolled copies.
Mitigations	Channel policy, classification matrix, certified-source requirement, ownership, lifecycle, monitoring and promotion path.
Implementation implications	Platform governance must maintain channel standards, connectors and support.
Cost implications	Licensing and operational cost across channels; reduce duplication and inactive assets.
Security implications	Classification and perimeter controls determine eligibility; BigQuery policies remain authoritative.
Operational implications	Central platform governs channels; domain/content owners govern assets.
Review triggers	Material adoption failure, control finding, product renaming/retirement or licensing change.
Review date	Semiannual, next formal review January 2027

ADR-ANA-003 - Use BigQuery reservations and workload assignments for critical analytics

Field	Decision record
ADR ID	ADR-ANA-003
Decision title	Use BigQuery reservations and workload assignments for critical analytics
Status	Approved with capacity validation
Date	11 July 2026
Architecture domain	Analytics performance and FinOps
Problem	Tier 0/1 dashboard workloads need predictable performance and protection from ad hoc analysis and engineering backfills.
Context	Global Fortune 100 analytics on inherited regional BigQuery data products and federated domain ownership.
Decision drivers	Concurrency; Predictability; Isolation; Cost attribution; Autoscaling; Operational simplicity

Field	Decision record
Constraints	Data residency, regulated workloads, thousands of users, existing BI coexistence and current product/licensing validation.
Assumptions	Certified data products exist; enterprise identity/groups and CI/CD foundations are available.
Alternatives considered	On-demand only; Single shared reservation; Dedicated capacity per workload/domain; Third-party acceleration
Evaluation criteria	Security, trust, scalability, usability, operability, cost, interoperability and reversibility.
Decision	Use reservations and assignments where measured contention or criticality justifies them. Protect Tier 0/1 BI from ad hoc and batch workloads. Derive baseline/autoscaling limits from tests and telemetry.
Rationale	BigQuery-native workload management preserves the inherited platform and enables isolation without separate warehouses.
Benefits	workload protection, attribution and capacity governance.
Tradeoffs	forecasting effort and risk of under/over-provisioning.
Positive consequences	Positive: workload protection, attribution and capacity governance.
Negative consequences	Negative: forecasting effort and risk of under/over-provisioning.
Risks	Incorrect assignment can starve workloads.; Capacity may be idle or autoscaling may exceed budget.
Mitigations	Load tests, utilization dashboards, shared idle-slot policy, budgets, staged changes and rollback.
Implementation implications	Consumer projects/folders and ownership metadata must support assignment; Terraform pipeline required.
Cost implications	Capacity cost based on edition, slots and autoscaling; use current pricing calculator, not fixed estimates.
Security implications	Reservation administration is separated from data access; no security boundary is assumed.
Operational implications	Platform/FinOps operate capacity; product owners define SLO and forecast.
Review triggers	Sustained low utilization, repeated queue breaches, edition change, cost variance or new region.
Review date	Monthly operational review; architecture review January 2027

ADR-ANA-004 - Enforce analytics security in BigQuery and add BI-layer controls

Field	Decision record
ADR ID	ADR-ANA-004
Decision title	Enforce analytics security in BigQuery and add BI-layer controls
Status	Approved
Date	11 July 2026
Architecture domain	Analytics security
Problem	BI-layer hidden fields, folders and model permissions alone cannot prevent direct or alternate-channel access to sensitive data.
Context	Global Fortune 100 analytics on inherited regional BigQuery data products and federated domain ownership.
Decision drivers	Zero Trust; Defense in depth; Multiple analytics channels; Auditability; Regulatory obligations
Constraints	Data residency, regulated workloads, thousands of users, existing BI coexistence and current product/licensing validation.
Assumptions	Certified data products exist; enterprise identity/groups and CI/CD foundations are available.
Alternatives considered	BI-only controls; BigQuery-only controls; Layered BigQuery plus BI controls
Evaluation criteria	Security, trust, scalability, usability, operability, cost, interoperability and reversibility.
Decision	Use BigQuery IAM, authorized views, row policies, policy tags and masking as authoritative controls. Use Looker roles, model sets, access filters and folders for persona and content governance.
Rationale	Layering preserves protection regardless of channel while improving user experience and least privilege.
Benefits	consistent protection and channel independence.
Tradeoffs	control coordination and more extensive testing.
Positive consequences	Positive: consistent protection and channel independence.
Negative consequences	Negative: control coordination and more extensive testing.
Risks	Policy mismatch may cause overexposure or broken content.; RLS can affect acceleration/performance.
Mitigations	Automated entitlement tests, metadata-driven policy, real-user performance testing, access reviews and exception expiry.
Implementation implications	Data classifications, groups and policy tags from prior volumes must be complete.
Cost implications	Security controls add engineering and possible performance cost but reduce breach/compliance risk.
Security implications	Primary security decision; supports VPC SC, CMEK and audit inheritance.

Field	Decision record
Operational implications	Security/governance own policy; data product and analytics teams implement and test.
Review triggers	New channel, new regulated dataset, policy engine change or critical access finding.
Review date	Quarterly and event-driven

19. Risks and Mitigations

ID	Risk	Probability	Impact	Mitigation / control owner
ANA-R01	Conflicting KPI definitions persist across tools.	High	High	KPI registry, certified measures, channel policy and dashboard inventory; Analytics Council.
ANA-R02	LookML model creates fanout or wrong totals.	Medium	High	Primary-key/join tests, reconciliation and peer review; Domain analytics lead.
ANA-R03	Critical dashboards contend with ad hoc queries.	High	High	Reservations, assignments, concurrency test and alerting; Platform/FinOps.
ANA-R04	BI-layer controls are mistaken for data security.	Medium	Critical	Authoritative BigQuery RLS/policy tags/masking and negative tests; Security/Data product.
ANA-R05	Data Studio/Sheets become shadow reporting systems.	High	High	Certified sources, publication tiers, owner/expiry, promotion and usage review; Analytics governance.
ANA-R06	Cross-region queries violate residency or create egress.	Medium	Critical	Regional models/connections, policy validation and aggregated approved products; Privacy/Architecture.
ANA-R07	BI Engine does not accelerate secured or complex queries.	Medium	Medium	Eligibility test with real personas, query redesign and standard BigQuery fallback; Performance team.
ANA-R08	Orphaned dashboards and schedules create risk/cost.	High	Medium	Group ownership, quarterly hygiene, inactivity retirement and schedule inventory; Domain lead.
ANA-R09	Embedded analytics exposes another tenant.	Low	Critical	Authoritative tenant filtering, server-side tokens, negative tests and security review; Application/Security.
ANA-R10	Regulatory report cannot be reproduced.	Low	Critical	Versioned code, cutoff, parameters, control totals, evidence retention and release freeze; Controller.
ANA-R11	Insufficient observability delays root cause.	Medium	High	Correlate Looker activity, BigQuery jobs, data freshness, release and cost; SRE.
ANA-R12	Capacity commitments exceed business value.	Medium	High	Variable cost model, staged capacity, shared idle slots, utilization review and rollback; FinOps.
ANA-R13	Product change invalidates documented capability.	Medium	Medium	Quarterly documentation validation, provider pinning and architecture review trigger; Platform architecture.
ANA-R14	Legacy BI coexistence creates duplicate cost and metrics.	High	High	Migration waves, authoritative report catalog, parallel validation and decommission criteria; Transformation lead.

20. Implementation Roadmap

Wave	Indicative duration	Objectives and deliverables	Dependencies	Exit criteria
0. Mobilize and assess	4-6 weeks	Inventory BI tools/reports/KPIs; classify use cases; decide instance/licensing; baseline query and cost.	Executive sponsorship, product/licensing input, prior-volume architecture.	Approved strategy, migration scope, owners and open-decision plan.
1. Platform foundation	6-10 weeks	Looker nonprod/prod topology, identity/groups, BigQuery connections, Git/CI, logging, initial capacity and guardrails.	Landing zone, networks, service perimeters, CI/CD and Terraform.	Security review, smoke tests, operational ownership and first connection.

Wave	Indicative duration	Objectives and deliverables	Dependencies	Exit criteria
2. Semantic MVP	8-12 weeks	Enterprise conventions, shared dimensions, 2-3 priority domain models, KPI registry and test framework.	Certified products and business owners.	Certified measures reconcile and representative Explores pass tests.
3. Executive and operational pilot	8-12 weeks	Tier 0/1 dashboards, BI Engine/reservation benchmark, SLOs, runbooks and support.	Semantic MVP, monitoring and control totals.	Load/security/recovery tests pass; users accept.
4. Governed self-service	8-16 weeks	Data Studio Pro, Connected Sheets and BigQuery Studio paved roads; content lifecycle; training and access workflow.	Channel licensing, Workspace/security and catalog integration.	Delegated users deliver within controls; no critical findings.
5. Scale and migrate	6-18 months	Migrate priority Tableau/Power BI/Qlik reports, rationalize KPIs, decommission duplicates and onboard domains.	Business migration waves and coexistence funding.	Authoritative catalog updated; legacy assets retired by criteria.
6. Optimize continuously	Ongoing	Capacity, performance, adoption, cost, content hygiene and automation improvements.	Operational telemetry and FinOps cadence.	SLOs met, unit cost stable/improving and exceptions reduced.

20.1 Migration decision pattern

Legacy asset outcome	Use when	Required action
Retire	Unused, duplicate or ownerless.	Validate no consumer, archive evidence, disable schedules and remove access.
Rebuild in Looker	Critical/reusable and benefits from governed semantic layer.	Map KPI, reconcile, performance-test, parallel run and cut over.
Retain temporarily	Contract, feature or timeline prevents migration.	Register owner, authoritative status, controls, cost and expiry decision.
Move to Data Studio Pro	Lightweight bounded report on certified source.	Remove local KPI logic, manage sharing and set review.
Replace with Connected Sheets	Spreadsheet analysis rather than managed dashboard.	Use live connection, approved source, owner and no broad distribution.
Replace with API/embed	Analytics must be integrated into workflow/application.	Application architecture, tenant security, load and support review.

21. Validation and Production Readiness

21.1 Architecture validation criteria

ID	Validation	Acceptance evidence
VAL-01	Every certified dashboard maps to certified data products and measures.	Catalog/lineage report and KPI IDs.
VAL-02	Regional connection/model boundaries match data residency.	Architecture diagram, connection inventory and negative test.
VAL-03	BigQuery controls protect data independent of BI content.	Direct query and alternate-channel security tests.
VAL-04	Looker roles, model sets and folders enforce least privilege.	Role matrix and persona tests.
VAL-05	Tier 0/1 performance meets SLO at 2x forecast peak.	Load and concurrency test.
VAL-06	Capacity assignments prevent ad hoc workload from exhausting critical BI.	Contention test and telemetry.
VAL-07	KPI changes are versioned, reviewed and reversible.	Git, CI/CD, approval and rollback evidence.
VAL-08	Critical reports are reproducible and reconciled.	Control package with data cutoff/code/report versions.
VAL-09	Data Studio Pro, Sheets and Studio follow channel policy.	Asset inventory, source/sharing/owner checks.
VAL-10	Monitoring correlates user experience, data freshness, query and cost.	Dashboard and sample incident trace.
VAL-11	Content ownership and lifecycle are active.	Orphan report and inactive schedule reports.
VAL-12	Current official product support, quotas and licensing have been validated.	Signed implementation validation record.

21.2 Production-readiness checklist

#	Production-readiness requirement	Status / evidence
1	Architecture decisions approved and deviations recorded.	Required before go-live
2	Looker edition, instance topology, locations, support and licensing approved.	Required before go-live
3	Production and nonproduction connections separated and tested.	Required before go-live
4	Identity groups, roles, model sets, content folders and privileged access reviewed.	Required before go-live
5	BigQuery RLS, policy tags, masking, authorized views and dataset IAM tested.	Required before go-live
6	VPC Service Controls/private connectivity paths validated, including conditional channels.	Required before go-live
7	Certified data products, KPI definitions, lineage and owners complete.	Required before go-live
8	LookML and semantic SQL repositories protected with code owners and CI.	Required before go-live
9	Model, data, reconciliation, content, security and performance tests pass.	Required before go-live
10	BigQuery reservations/assignments/BI Engine sized and load-tested where used.	Required before go-live
11	SLOs, alerts, on-call routes, runbooks and error budgets active.	Required before go-live
12	Backup/reconstruction and rollback procedures tested.	Required before go-live
13	Billing attribution, budgets, anomaly detection and optimization cadence active.	Required before go-live
14	Dashboard and report standards applied; accessibility and sharing checked.	Required before go-live
15	Regulatory/critical report evidence, retention and release freeze defined.	Required before go-live
16	Content inventory, ownership, expiry and orphan cleanup implemented.	Required before go-live
17	Vendor escalation, quota management and service limits documented.	Required before go-live
18	Training, support and access-request workflows available.	Required before go-live

22. Dependencies, Open Decisions and Next Actions

Type	Item	Owner / target
Dependency	Certified BigQuery data products, data contracts, quality and lineage from Volumes 03-05.	Data product owners / before model build.
Dependency	Landing-zone identity, network, logging, KMS and VPC SC controls.	Cloud platform / before production connection.
Dependency	Enterprise CI/CD and Terraform workflows.	DevSecOps / before production promotion.
Open decision	Looker edition, instance count, locations and recovery topology.	CIO/CDO/Architecture / Wave 0.
Open decision	Coexistence and migration plan for Tableau, Power BI and Qlik.	Analytics transformation / Wave 0.
Open decision	BigQuery capacity baseline, autoscaling and chargeback method.	FinOps/platform / pilot load test.
Open decision	Data Studio Pro and Connected Sheets eligibility for Confidential data.	Security/privacy / Wave 4.
Open decision	Regulatory report evidence repository and approval workflow.	Compliance/records / pilot.
Next action	Approve ADR-ANA-001 through ADR-ANA-004.	Architecture Review Board.
Next action	Select two priority domains and one executive KPI family for the MVP.	CDO and business owners.
Next action	Run current official product, quota, region, license and provider validation.	Platform architecture.
Next action	Build nonproduction Looker/BigQuery path and execute a real-user security/performance proof.	Platform, security and domain team.

23. Appendices and References

Appendix A - Artifact Inventory

Artifact	Location / purpose
Detailed guide	documents/Volume_06_Analytics_Platform_Detailed.docx and .pdf
Architecture diagrams	diagrams/ in PNG, SVG, Mermaid, PlantUML, Graphviz DOT and draw.io formats
LookML example	examples/looker/
BigQuery SQL examples	examples/sql/
Terraform capacity example	examples/terraform/
Analytics access matrix	matrices/analytics_access_matrix.csv
KPI definition template	matrices/kpi_definition_template.csv
Package manifest	MANIFEST.txt
Package readme	README.md

Appendix B - Diagram Construction Specification

Each draw.io file is directly editable. For enterprise icon refinement, retain the logical structure below and replace generic nodes with current official Google Cloud product icons. Use swimlanes for Consumer, Semantic, Data Product, Acceleration and Control planes. Show AMER, EMEA and APAC boundaries, VPC Service Controls perimeters for restricted data, environment boundaries, data-flow arrows and control-flow arrows. Use separate views instead of adding every component to one diagram.

Diagram	Required boundaries and connectors
Analytics platform	Certified products -> semantic layer -> channels; security/governance and operations cross-cutting.
Executive dashboard	Domain products -> enterprise KPI model -> Looker delivery -> decision forums, with assurance flow.
KPI lifecycle	Definition -> design -> validation -> certification -> publication -> monitoring -> change/retire.
Security/access	IdP -> groups -> Looker and BigQuery controls -> protected products -> audit.
Delivery/operations	Git -> CI -> promotion -> runtime -> observability -> rollback.

Appendix C - Official Documentation Validation Set

- Looker documentation - validate current features, regions, quotas, IAM and commercial terms before implementation.
- LookML model development - validate current features, regions, quotas, IAM and commercial terms before implementation.
- Looker roles and access - validate current features, regions, quotas, IAM and commercial terms before implementation.
- Looker CI and validation - validate current features, regions, quotas, IAM and commercial terms before implementation.
- Looker embedding - validate current features, regions, quotas, IAM and commercial terms before implementation.
- BigQuery documentation - validate current features, regions, quotas, IAM and commercial terms before implementation.
- BigQuery BI Engine - validate current features, regions, quotas, IAM and commercial terms before implementation.
- BigQuery cached results - validate current features, regions, quotas, IAM and commercial terms before implementation.
- BigQuery reservations - validate current features, regions, quotas, IAM and commercial terms before implementation.
- BigQuery materialized views - validate current features, regions, quotas, IAM and commercial terms before implementation.
- BigQuery Studio - validate current features, regions, quotas, IAM and commercial terms before implementation.
- Connected Sheets - validate current features, regions, quotas, IAM and commercial terms before implementation.
- Data Studio Pro - validate current features, regions, quotas, IAM and commercial terms before implementation.
- Terraform Google provider - validate current features, regions, quotas, IAM and commercial terms before implementation.

Document conclusion. Volume 06 establishes the enterprise analytics consumption and semantic architecture without changing Volumes 01-05. Its approval authorizes implementation planning and pilots, subject to the stated open decisions and current-product validation.